

Physics for Quantum Computing

Lecture 1: Introduction – Why Physics Matters?

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Course Learning Outcomes

- Understand why quantum mechanics is essential for modern computing.
- Appreciate the limitations of classical computation.
- Relate quantum physics to quantum information processing.
- Obtain an overview of the course structure.

Why Study Quantum Computing?

Motivation

Nature behaves according to the laws of quantum mechanics. Future computers exploit these laws instead of fighting them.

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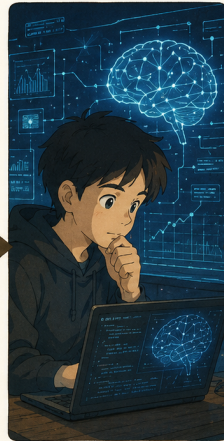
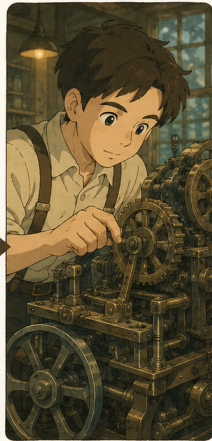
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Examples:

- Drug discovery
- Materials design
- Cryptography
- Optimization

Evolution of Computing



Abacus → Mechanical → Electronic → AI → Quantum

Why Classical Computing Reaches a Limit

- Transistors are approaching atomic dimensions.
- Electrons exhibit quantum tunnelling.
- Classical physics fails to describe nanoscale devices.
- A new computational paradigm is required.

Historical Motivation

Important experimental observations:

- ① Blackbody radiation
- ② Photoelectric effect
- ③ Atomic spectra
- ④ Wave–particle duality

These observations led to quantum mechanics.

Course Roadmap

- ① Light–Matter Interaction
- ② Quantum Mechanics
- ③ Qubits
- ④ Quantum Gates
- ⑤ Quantum Circuits
- ⑥ Quantum Algorithms
- ⑦ Quantum Hardware

How Physics Leads to Quantum Computing

Atoms $\rightarrow$ Quantum States $\rightarrow$ Superposition $\rightarrow$ Qubits $\rightarrow$ Quantum Gates $\rightarrow$ Algorithms

Summary

- Quantum computing originates from quantum physics.
- Classical computers have physical limits.
- Understanding quantum mechanics is essential before studying qubits.
- Next Lecture: Light–Matter Interaction.